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$\boxed{\text{Generative methods}}$

Explicit

$$\boxed{p_\theta(x) \sim \frac{e^{-E(x;\theta)}}{Z(\theta)}}$$

$\boxed{\text{Tractable}}$ → $\boxed{\text{Normalizing flows}}$

$\boxed{\text{approx}}$ → $\boxed{\text{BM}}$ $\boxed{\text{VAE}}$ $\boxed{\text{DIFF}}$

$$\nabla_\theta \log p_\theta(x)$$

Implicit

$\boxed{\text{GANs}}$

if $z \sim P_Z(z)$

(normally known)

$x = f(z)$ where f is an invertible function

we can then define a PDF for x

$$P_X(x) = P_Z(f^{-1}(x)) \left| \det \frac{\partial f^{-1}}{\partial x} \right|$$

$x = f(z)$ if invertible

$z = f^{-1}(x)$

Probability is assumed to be conserved

$$P_x(x)\,dx = P_z(z)\,dz$$

in one dimension, since

$z = f^{-1}(x)$

$$P_x(x) = P_z\left(f^{-1}(x)\right)\left|\frac{dz}{dx}\right|$$

$$p_X(x) = p_Z\left(f^{-1}(x)\right) \left| \frac{d f^{-1}(x)}{dx} \right|$$

in several dimensions

$$p_X(x) = p_Z\left(f^{-1}(x)\right) \left| \det \frac{\partial f^{-1}}{\partial x} \right|$$

equivalently, since

$$\frac{\partial f^{-1}}{\partial x} = \left(\frac{\partial f}{\partial z} \right)^{-1}$$

$$p_X(x) = p_Z(z) \left| \det \frac{\partial f}{\partial z} \right|^{-1}$$

$$P_z(z)\,dz = \begin{cases} dz & z \in [0,1] \\ 0 & \text{else} \end{cases}$$

$z \in [0,1]$

$$P_X(x)\,dx = e^{-x}\,dx$$

$x \in [0,\infty)$

what is $f(x)$ and $f^{-1}(z)$

$$P_z(z)\,dz = dz = P_X(x)\,dx$$

$$= e^{-x}\,dx$$

$$z(x) = f(x) = \int_0^x P_X(x')\,dx'$$

(cumulative dist $P(x)$)

$$z(x) = P(x) = \int_0^x e^{-x'}\,dx'$$

$$= 1 - \exp(x)$$

The inverse gives

$$x(z) = -\ln(1-z)$$

$z=1 \Rightarrow x = -\ln 0 = +\infty$

$z=0 \Rightarrow \ln 1 = 0$